

Artifact 25 — Synthetic PHI Generator

Execution · Stage C · Validation Environment

Instruction to ECC: "Build this feature so it passes the tests in incidents.json, including the SPP decay scenarios (SYN-04). All 62 ground truth events in expected_audit.sql must be produced with correct structure. The SD-01B merge gate query must return 0 rows."

DOCUMENT DETAILS

Stage	Execution Plugin — Stage C (Validation Environment)	Skill	synthetic-phi-generator
Date	2026-03-28	Outputs	clients.json, incidents.json, expected_audit.sql
Input	Artifact 23 (Agentic Logic Spec) § 3 Threshold Constants + §8 Canonical Event Registry + §9 Integration Test Scenarios		

Table of Contents

Table of Contents	2
Privacy Contract	4
File Inventory	5
Test Scenario Summary	6
Scoring Validation Reference	7
Guard and Compliance Events	8
Model Selection Compliance	9
Hash Chain Requirements	10
Usage Instructions	11
Step 1 — Load test fixtures into local database	11
Step 2 — Run ECC against test scenarios	11
Step 3 — Run the SD-01B merge gate query	11
Step 4 — Check the detection accuracy summary	11
Step 5 — Interpret results	11
Step 6 — Hand off to harness-audit-grader (Artifact 26)	12
Known Pre-Sprint Gate Dependencies	13
Implementation Stub Resolutions (Amendment 3-A / 3-B / 3-C)	14
Gap 1 — SMS Template Copy Approval (B1-B)	14
Gap 2 — Google Maps / IS-03 Postcode Fallback (SC-07)	14
Gap 3 — AX-02 Trigger Source Guard (AlayaCare Write-Back)	15
New Files Created by Amendments 4-A and 5-A	17
How config/sms_templates.py moves from advisory to binding	17

Privacy Contract

All data in these files is entirely fictitious. No real client, carer, or agency data was used or referenced. All identifiers are deterministic UUIDs chosen for test reproducibility, not randomly generated. Data is statistically plausible for Australian home care operations but non-attributable.

Regulatory frame: Australian Privacy Act 1988 (APP). HIPAA architecture applied as design floor only. Cross-border disclosure (APP 8): `WHATSAPP_APP8_CONFIRMED=false` and `SC07_GOOGLE_MAPS_APPROVED=false` — both features remain blocked until legal confirmation.

P-9 absence: The P-9 SPP field (disability/medical condition detail) is explicitly absent from all records per AX-01 pre-sprint gate. It must not appear in any schema, audit log, or matching logic.

File Inventory

File	Analogous to (Skill Template)	Purpose
synthetic-data/clients.json	patients.json	5 fictitious client SPP profiles + 9 carer roster records
synthetic-data/incidents.json	sensors.json	6 vacancy incident scenario traces with expected state transitions
synthetic-data/expected_audit.sql	alerts.sql	Ground truth APPAuditLogEntry events (62 total) + SD-01B merge gate query

Test Scenario Summary

Scenario	Client	Event Type	Injection Window	Expected Terminal State	Key Assertion	Detection SLA
SYN-01	Margaret Chen (cli-0001)	Familiarity ranking — happy path	2026-04-15 07:30	RESOLVED	David Okafor rank 1; composite=0.726; phrasing='who has visited you before'	30 sec
SYN-02	Patricia Williams (cli-0002)	All 3 carers decline + HITL double timeout	2026-04-16 08:00	VACANCY_UNRESOLVED	3× CARER_DECLINED; Sonnet invoked; no auto-assign; alert_pm_lead	30 sec
SYN-03	Dorothy Kumar (cli-0003)	KNOWN_CARERS_ONLY + empty P-7 → empty shortlist	2026-04-17 12:00	VACANCY_UNRESOLVED	shortlist_size=0; G-DS-05 guard_passed=false; no carer contacted	30 sec
SYN-04a	Evelyn Nguyen (cli-0004)	SPP T=0 baseline — familiarity active	2026-04-18 06:00	RESOLVED	Michael Santos rank 1; composite=0.622; no SPP_COLD_START event	30 sec
SYN-04b	Evelyn Nguyen (cli-0004)	SPP decay T+30 → proximity-only	2026-05-18 06:00	RESOLVED	SPP_COLD_START fires; Emma Davies now rank 1; ranking change confirmed	30 sec
SYN-05	Jean Morrison (cli-0005)	New client EC-04 → MINIMAL briefing	2026-04-19 08:00	RESOLVED	SPP_COLD_START; briefing_mode=MINIMAL; spp_invitation_url on card	30 sec

Scoring Validation Reference

All composite scores use constants from Artifact 23 §3:

```
> SCORING_WEIGHT_FAMILIARITY = 0.60
> SCORING_WEIGHT_PROXIMITY = 0.40
> PROXIMITY_MAX_KM = 25
> familiarity_score = min(visit_count × 0.20, 0.60)
> proximity_score = 1 - (distance_km / PROXIMITY_MAX_KM)
> composite_score = (familiarity_score × 0.60) + (proximity_score × 0.40)
```

Scenario	Carer	Distance	Familiarity	Proximity	Composite	Rank
SYN-01	David Okafor	2.11 km	0.60 (3 visits)	0.916	0.726	#1
SYN-01	Thomas Patel	2.56 km	0.00	0.898	0.359	#2
SYN-01	Sarah Brennan	3.56 km	0.00	0.858	0.343	#3
SYN-02	Robert Kim	2.22 km	0.00	0.911	0.364	#1
SYN-02	Lisa Wong	2.53 km	0.00	0.899	0.360	#2
SYN-02	James Sullivan	2.71 km	0.00	0.892	0.357	#3
SYN-04a	Michael Santos	1.13 km	0.40 (2 visits)	0.955	0.622	#1
SYN-04a	Emma Davies	0.33 km	0.00	0.987	0.395	#2
SYN-04a	Anna Fitzgerald	1.94 km	0.00	0.922	0.369	#3
SYN-04b	Emma Davies	0.33 km	0.00 (SPP null)	0.987	0.395	#1 ← ranking change
SYN-04b	Michael Santos	1.13 km	0.00 (SPP null)	0.955	0.382	#2 ← demoted
SYN-04b	Anna Fitzgerald	1.94 km	0.00	0.922	0.369	#3
SYN-05	Anna Fitzgerald	0.50 km	0.00	0.980	0.392	#1
SYN-05	Emma Davies	2.41 km	0.00	0.904	0.362	#2
SYN-05	Michael Santos	3.28 km	0.00	0.869	0.348	#3

SYN-04 ranking change is the primary SPP decay test: Michael Santos has a genuine visit history (P-7={ca3e0005: 2}) but when P-10=0.0 his familiarity advantage is suppressed. Emma Davies, who is 0.33 km from Evelyn, takes the top spot purely on proximity.

Guard and Compliance Events

Every scenario exercises one or more compliance guards. ECC must log these guards correctly:

Guard ID	Event Type Logged	Scenario(s)	guard_passed	Critical Assertion
CC-8	CC8_FIELD_STRIPPED	All (every CARER_NOTIFIED preceded by this)	TRUE	PHI fields stripped from SMS body before send
CC-6	BRIEFING_SENT	All (briefing events)	TRUE	No match explanation in briefing content
G-DS-05	SHORTLIST_GENERATED	SYN-01 (pass), SYN-03 (block)	TRUE / FALSE	SYN-01: familiarity phrasing applied. SYN-03: guard_passed=FALSE, block_reason contains 'KNOWN_CARERS_ONLY'
G-E3-1	CLIENT_NOTIFIED, FAMILY_NOTIFIED	SYN-01, 04a, 04b, 05	TRUE	CLIENT_NOTIFIED event must precede FAMILY_NOTIFIED in every resolved scenario
CC-4	(not an event — field exclusion)	All scenarios	n/a	P-2/P-4/P-5 fields must never appear in matching_criteria or any scored output

Model Selection Compliance

Per Artifact 23 §3 and HS-STRAT-03:

Event	Required Model	Scenarios
VACANCY_DETECTED, SHORTLIST_GENERATED, HITL_REQUESTED	NO_LLM or claude-haiku-4-5-20251001	All
SPP_COLD_START, SHORTLIST_GENERATED (scored)	claude-haiku-4-5-20251001	SYN-04b, SYN-05
HITL_DOUBLE_TIMEOUT, VACANCY_UNRESOLVED	claude-sonnet-4-6	SYN-02, SYN-03
BRIEFING_SENT, CLIENT_NOTIFIED, FAMILY_NOTIFIED, CC8_FIELD_STRIPPED	NO_LLM	All

Critical: `syn02-aud-0017` (HITL_DOUBLE_TIMEOUT) and `syn03-aud-0003` (VACANCY_UNRESOLVED) must record `model_id='claude-sonnet-4-6'`. Token budget must not exceed `TOKEN_BUDGET_L3=3500`. ECC must assert this at runtime.

Hash Chain Requirements

Every APPAuditLogEntry must carry:

- entry_hash: SHA-256(CONCAT(all_field_values_above_this_field))
- previous_hash: entry_hash value of the immediately preceding entry in the same case_id
- First entry in a case uses genesis sentinel: 'CHAIN_GENESIS_{scenario}' (as per expected_audit.sql)

The SD-01B merge gate query checks hash_chain_valid in synthetic_detected_audit_events. A broken hash chain is a CRITICAL finding that blocks merge.

Usage Instructions

Step 1 Load test fixtures into local database

```
# PostgreSQL (recommended for production-parity testing)
psql -d homecare_test -f Execution/synthetic-data/expected_audit.sql

# Or SQLite for CI
sqlite3 homecare_test.db < Execution/synthetic-data/expected_audit.sql
```

Step 2 Run ECC against test scenarios

```
# Point the matching engine at the synthetic client/carer data
export HOMECARE_AI_TEST_MODE=true
export HOMECARE_AI_CLIENTS_FIXTURE=Execution/synthetic-data/clients.json
export HOMECARE_AI_INCIDENTS_FIXTURE=Execution/synthetic-data/incidents.json

# Execute all 6 incident scenarios in sequence
python -m homecare_ai.test_harness --scenarios SYN-01,SYN-02,SYN-03,SYN-04a,SYN-04b,SYN-05
```

ECC reads each incident from `incidents.json`, simulates the stimulus events (vacancy trigger, carer declines, HITL timeouts) in the order specified by `event_timestamps`, and writes `APPAuditLogEntry` records to `synthetic_detected_audit_events`.

Step 3 Run the SD-01B merge gate query

```
-- Run against your test database after ECC populates detected_audit_events
\i Execution/synthetic-data/expected_audit.sql -- loads DDL + ground truth

-- Then run the merge gate check:
-- (The query is included at the bottom of expected_audit.sql)
-- Expected result: 0 rows returned = PASS
```

Step 4 Check the detection accuracy summary

Run the first validation query (also at the bottom of `expected_audit.sql`). Expected pass criteria:

Metric	Target
detection_rate_pct	100% for all scenarios
phi_violations_CRITICAL	0 (any value > 0 = hard block)
hash_chain_valid	100%
model_id_correct	100% (especially SYN-02 HITL_DOUBLE_TIMEOUT = Sonnet)
guard_id_correct	100% for CC-8, G-DS-05, CC-6, G-E3-1 events
missing_non_nullable_fields	0

Step 5 Interpret results

Merge Gate Result	Action
0 rows — PASS	Eligible for code review and production deploy gate
Rows returned	Review defect_type column; create Agent Fix Queue items (Artifact 26); fix and re-run from Step 2

Step 6**Hand off to harness-audit-grader (Artifact 26)**

After SD-01B PASS: 1. Export detected audit events: `COPY synthetic_detected_audit_events TO 'detected_audit_export.csv'` 2. Pass `clients.json`, `incidents.json`, `expected_audit.sql`, `detected_audit_export.csv`, and ECC source code to `harness-audit-grader` 3. Grader runs Security Debt (8 check groups) and Requirement Drift (6 check groups) checks 4. Fix any CRITICAL or HIGH findings; re-run until `OVERALL VERDICT = PASS`

Known Pre-Sprint Gate Dependencies

These items must be resolved before any real-data test run. Synthetic data tests are not affected.

Gate	Status	Due	Impact
AX-01	Pending engineer confirmation	Before Sprint 1	P-9 field absent from schema
AX-02	Result due 2026-04-01	2026-04-01	If write-back unavailable → US-14 (Manual Absence) enters Sprint 2
SC-07	Legal review pending	2026-04-01	Google Maps DPA vs APP 8; postcode fallback IS-03 active until confirmed
Copy Approval (B1-B)	Pending Privacy Counsel	Before Sprint 1	All 4 SMS templates require APP compliance sign-off
Sketch-gate	Pending designer	Before Sprint 1	US-10 coordinator card wireframe required

Implementation Stub Resolutions (Amendment 3-A / 3-B / 3-C)

Three gaps identified post-generation. Each is resolved below with a binding status.

Gap 1 SMS Template Copy Approval (B1-B)

RISK

The 4 SMS templates (ACT-C-01, ACT-C-02, ACT-P-01, ACT-F-01) are verbatim in Artifact 23 §4.1 but not yet reviewed by Privacy Counsel. Synthetic tests pass on template-name assertions only. A test passing on placeholder text is not evidence of APP compliance.

RESOLUTION

All SMS body content assertions in `incidents.json` are marked **ADVISORY** until B1-B is cleared. The binding gate is not in the test harness — it is in `config/sms_templates.py`:

```
# config/sms_templates.py – these fields MUST be non-null before Sprint 1 start
SMS_TEMPLATES_VERSION = "1.0.0"
SMS_TEMPLATES_APPROVED_BY = None # ← BLOCKS go-live until Privacy Counsel signs
SMS_TEMPLATES_APPROVED_DATE = None # ← BLOCKS go-live until Privacy Counsel signs
```

harness-audit-grader (Artifact 26) check group **CP-01** must assert:

- `config/sms_templates.py.approved_by` IS NOT NULL
- `config/sms_templates.py.approved_date` IS NOT NULL

If either is null: CP-01 finding = **HIGH** (not CRITICAL — synthetic tests can still run; go-live is blocked).

What ECC must not do: Treat a passing test as proof that the SMS text is APP-compliant. The only proof is Privacy Counsel sign-off in approved_by.

Gap 2 Google Maps / IS-03 Postcode Fallback (SC-07)

RISK

SC07_GOOGLE_MAPS_APPROVED=false. All 6 incidents include client/carer lat/lng coordinates derived from ABS postcode centroids (`data/au_postcode_centroids.csv`). If ECC calls the Google Maps API to resolve coordinates instead of reading the centroid file, it sends postcodes to a third-party US-hosted service without an APP 8 cross-border disclosure basis — a **CRITICAL** data sovereignty violation.

RESOLUTION

Two binding assertions are now encoded in the test files:

1. `incidents.json` `global_validation_assertions[GLOBAL-IS03]`: Every `SHORTLIST_GENERATED` audit entry must contain `IS-03:POSTCODE_CENTROID_FALLBACK` in `threshold_applied` when `SC07_GOOGLE_MAPS_APPROVED=false`.
2. `expected_audit.sql` **SHORTLIST_GENERATED events**: All 6 `threshold_applied` fields now include `IS-03:POSTCODE_CENTROID_FALLBACK`. If ECC uses the Maps API, the centroid flag will be absent and the SD-01B merge gate extension query will fire.

What ECC must implement

```
# In matching_engine.py – proximity computation
if not SC07_GOOGLE_MAPS_APPROVED:
```

```
# IS-03: Use ABS postcode centroids – never call Maps API
centroid = lookup_centroid(postcode, "data/au_postcode_centroids_seed.csv")
distance_km = haversine(client.lat, client.lng, centroid.lat, centroid.lng)
audit_entry.threshold_applied += " ,IS-03:POSTCODE_CENTROID_FALLBACK"
else:
    # SC-07 confirmed – Maps API permitted
    distance_km = google_maps_distance(client_address, carer_address)
```

Data file — now exists in repo

data/au_postcode_centroids_seed.csv — 13 rows covering all postcodes in the 6 test scenarios. ECC can run the full harness against this seed file without the full ABS dataset.

For production: replace with the full ABS Postal Areas ASGS Edition 3 file (~2.8 MB, ~18,000 rows). Download URL is in the seed file header comment. Pre-load into Lambda /tmp on cold start.

Merge gate: `expected_audit.sql` IS-03 guard query (bottom of file) — must return 0 rows.

Gap 3 AX-02 Trigger Source Guard (AlayaCare Write-Back)

RISK

All 6 incidents specify `trigger_source: "alayacare_absence_api"`. AX-02 result is due 2026-04-01. If the AlayaCare write-back is unavailable, the vacancy detection trigger changes to `MANUAL_COORDINATOR_ENTRY` (US-14). Running the test harness against `alayacare_absence_api` when write-back is unavailable produces a **false pass** — the feature appears to work but the integration path is wrong.

RESOLUTION

The test harness must read `HEMOCARE_AI_AX02_STATUS` before executing any scenario:

```
# Required env var – must be set before running test harness
export HEMOCARE_AI_AX02_STATUS=CONFIRMED # AlayaCare write-back available
# OR
export HEMOCARE_AI_AX02_STATUS=UNAVAILABLE # Write-back unavailable → US-14 path
```

Harness behaviour by AX-02 status

AX02_STATUS	Expected VACANCY_DETECTED.trigger_sensor	What changes
CONFIRMED	alayacare_absence_api	All incidents run as written
UNAVAILABLE	MANUAL_COORDINATOR_ENTRY	Harness substitutes trigger_source; US-14 flow tested instead
unset	—	Harness aborts with exit code 2: AX02_NOT_CONFIRMED

```
# test_harness.py – pre-test abort gate
ax02_status = os.environ.get("HEMOCARE_AI_AX02_STATUS")
if ax02_status not in ("CONFIRMED", "UNAVAILABLE"):
    sys.exit(2) # AX02_NOT_CONFIRMED – ambiguous test state; do not run
```

If `AX02_STATUS=UNAVAILABLE`: US-14 (Manual Absence Record) stories enter Sprint 2. SYN-01 through SYN-05 `vacancy_details.trigger_source` is overridden to `manual_coordinator_entry` at harness runtime. The `VACANCY_DETECTED` audit entry must then log `trigger_sensor='MANUAL_COORDINATOR_ENTRY'` per US-14 AC-2 (Artifact 24).

Merge gate: `expected_audit.sql` AX-02 false-pass guard query (bottom of file) — must return 0 rows. A mismatch between `trigger_sensor` in the detected audit entry and the expected value for the confirmed `AX02_STATUS` is a CRITICAL finding.

New Files Created by Amendments 4-A and 5-A

File	Lines	Created By	Purpose
[data/au_postcode_centroids_seed.csv](../data/au_postcode_centroids_seed.csv)	14 + header	Amendment 4-A	IS-03 lookup data for 13 test postcodes; replace with full ABS file for production
[config/sms_templates.py](../config/sms_templates.py)	~190	Amendment 5-A	Verbatim SMS templates (all 4 actions, 8 variants); approved_by=None blocks production; test harness bypasses in advisory mode

How config/sms_templates.py moves from advisory to binding

Now:

```
APPROVED_BY = None → harness runs in advisory mode
APPROVED_DATE = None → harness runs in advisory mode
harness-audit-grader CP-01 → ADVISORY (non-blocking)
```

After Privacy Counsel review (B1-B):

```
APPROVED_BY = "Privacy Counsel – [Name]"
APPROVED_DATE = "2026-XX-XX"
harness-audit-grader CP-01 → PASS (binding)
incidents.json SYN01-A9 through SYN01-A12 → advisory flag removed → BINDING assertions
```

No code changes required — only the two fields in config/sms_templates.py change. All template strings are already final (verbatim from Artifact 23 §4.1). ECC must not inline these strings anywhere else in the codebase.

Artifact 25 is the final Stage C artifact. Stage D begins: hand PRD (Artifact 21) + Logic Spec (Artifact 23) + User Stories (Artifact 24) + Synthetic Data (Artifact 25) to ECC. Run harness-audit-grader (Artifact 26) until OVERALL VERDICT = PASS.